VIVEK SHUKLA

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GitHub | Portfolio | LinkedIn

SUMMARY

MCA student with a strong foundation in **mathematics, statistics, data analysis, and programming**, with hands-on experience in **insurance risk analytics, quantitative analysis, and data validation**. Built an insurance analytics system covering **claim frequency, claim severity, pure premium, loss ratio, earned premium, and rate adequacy**. Proficient in **Python, SQL, Excel, pandas, and NumPy**, with an interest in applying quantitative methods and financial modelling to insurance risk and actuarial problems.

EDUCATION

Jawaharlal Nehru University (JNU), New Delhi 2025 – 2027
Master of Computer Applications (MCA) *CGPA: 7.54 / 9*
Relevant coursework: Statistics

Siddharth University, Kapilvastu 2021 – 2024
Bachelor of Computer Applications (BCA) *CGPA: 8.12 / 10*
Relevant coursework: Statistics

UP Board 2021
Class XII – Mathematics: 84%, English: 81% *Overall: 83.40%*

TECHNICAL SKILLS

- **Risk & Quantitative Analysis:** Risk Analysis, Statistical Analysis, Quantitative Analysis, Scenario Analysis, Risk Segmentation, Financial Data Analysis, Financial Modelling Concepts
- **Insurance Analytics:** Claim Frequency, Claim Severity, Pure Premium, Loss Ratio, Earned Premium, Rate Adequacy
- **Data Analysis:** Python, pandas, NumPy, SQL, MySQL, Data Validation, Data Cleaning, Data Reconciliation, Data Quality Checks
- **Spreadsheets & ML:** MS Excel, Pivot Tables, VLOOKUP, MIS Reporting; Text Classification, TF-IDF, NLTK

PROJECTS

Insurance Risk & Premium Analytics System GitHub
React, FastAPI, Python, pandas, NumPy, MongoDB Atlas
- Built a full-stack insurance analytics system to analyze synthetic policy and claims datasets across **portfolio analysis, claims analysis, risk segmentation, scenario analysis, and rate adequacy**.
- Implemented quantitative insurance calculations for **claim frequency, claim severity, pure premium, loss ratio, and earned premium** using Python, pandas, and NumPy.
- Developed **risk and premium analysis** across policy type, age band, and gender segments using credibility-weighted rate indication and configurable target loss ratios.
- Implemented **valuation-date-aware calculations, scenario validation, filter-scope checks, and edge-case handling** to improve consistency and reliability of risk and financial analysis.
- Added regression tests for formulas and edge cases; **127 backend tests passed (1 skipped), 10 frontend tests passed**.

Spam Detection System GitHub
Python, NLTK, TF-IDF, Machine Learning, Streamlit, Tkinter
- Built a machine learning system to classify **spam and legitimate SMS/email messages**, using a preprocessing workflow (tokenization, stopword removal, stemming), **TF-IDF feature extraction**, and Streamlit/Tkinter interfaces for real-time prediction.

EXPERIENCE

People Action For National Integration (PANI) Jul 2025 – Oct 2025
MIS Assistant – Utraula CSAL-2 Project
- Validated and audited **11,000+ farmer records** using Excel and MIS workflows, identifying missing, inconsistent, and incorrect information before reporting.
- Performed structured data analysis across field submissions and communicated with stakeholders including **28 CRPs/project teams** and farmers to investigate discrepancies, verify information, and maintain reliable records.
- Prepared and presented **monthly MIS reports and quantitative data summaries** using sorting, filtering, Pivot Tables, VLOOKUP, and data reconciliation techniques.
- Reviewed current and historical project data to identify **inconsistencies, reporting gaps, and data-quality issues**, followed up for corrections, and maintained updated datasets for analysis.

ACHIEVEMENTS

- Ranked in the **Top 30% among 2,989 participants** in the OpenAI Codex Hackathon, organized by NamasteDev.